Data Setup Automation Design

VMM PO Governance — Data Setup Automation: Design

Status: proposal for review · **Date:** 27 Oct 2026 · **Applies to:** POGovApi / Admin portal (this repository)

Decision asked for: replace the daily manual upload in 📂 Data Setup with an automatic import: SAP BW writes the data files every night (BW Open Hub → CSV, 04:00), the tool loads them by itself, checks them against a control file, and keeps the previous data if anything is wrong. Manual upload stays as the fallback. In a later phase, SAP's live PO / goods-receipt / return events (already received on `POST /api/sap/v1/events`) keep the open-PO and stock position current during the day.

Nothing in the decision rules changes. The engines keep reading the same datasets (F1…F4, Universe Master, BSR, masters); only the way the datasets arrive changes.

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1. Summary

	Today	**After phase 1 (BW feed)**	**After phase 3 (live position)**
Who loads the data	A planner, by hand, every morning	Nobody — the tool, automatically at ~04:30	Same, plus SAP events during the day
How	Excel / CSV / ZIP upload in Data Setup	BW Open Hub CSV files + control file in a landing folder	+ `POST /events` applied to a live open-PO / stock position
Freshness	Whenever someone uploads (often mid-morning)	Previous day's close, ready before office hours	Near real time for POs, receipts and returns
Checks	Visual; row counts in the upload report	Control file: row count + total qty per file, master checks, held-aside rows	+ event order / cut-off checks
If data is missing	Decisions run on yesterday's (or older) upload	Previous base is kept, BW team + planning are emailed, `data_delayed` can be raised	Same
SAP `bw_snapshot`	Upload time of F1	BW snapshot time from the control file	Same

Planned effort (tool side): **phase 1 about 5–6 weeks**, **phase 2 (plan + remaining masters) about 2 weeks**, **phase 3 about 6–8 weeks** after the open questions in §12 are answered. BW and SAP work is on top (§11).

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2. How Data Setup works today

2.1 The datasets

The tool stores eleven datasets. Each upload creates a new **version** (gzip JSON in `dataset_versions`); the **manifest** says which version is current, and the engines read an in-memory snapshot of the current versions.

Key	Name on screen	What it is	Needed by	Upload
`f1`	F1 — Stock	Closing stock by Brand × MC × MRP × Flag × Colour (+ GM), company level	**Tool 1, SAP decisions**, Tool 2, Recommendation, Attribute Analysis	single
`f2`	F2 — Pending PO	Released open PO lines: PO no, GM, colour, delivery month, pending qty, value, vendor	**Tool 1, SAP, Tool 2** (its line list), Recommendation	single
`f3`	F3 — Sales Trend	Daily sales of the last 45 days by colour (`Day 1` = oldest)	**Tool 1, SAP**, Tool 2, Recommendation, Attribute Analysis	single
`f4`	F4 — Forecast	Monthly plan by Brand × MC × MRP × Flag (`MM.YYYY` columns)	**Tool 1, SAP**, Tool 2, Recommendation	single
`um`	Universe Master	GMs of the Single Concept MRP universe (routes a GM to engine B)	**Tool 1, SAP**, Tool 2, GRC	single
`bsr`	BSR — All Seasons	Per GM + colour: weeks on floor, average per-day sell-through %	**Tool 1, SAP** (engine A), Recommendation, GRC	**multi** (files accumulate)
`artmaster`	Article Master	GM → brand, division, MC, MRP, flag, season, style, fit, material, design, vendor	Resolve map (top priority), seasons, Recommendation A, Attribute Analysis	single
`colmaster`	Colour Master	Colour code → description	Tool 1 colour list, SAP colour matching	single
`attrrel`	Attr Relevance Master	Brand × MC × MRP × Flag → assessment dimension (Style / Fit / Material / Design)	Recommendation A	single
`catheadmap`	SCR Products / Cat Head Mapping	Brand × MC × MRP → buyer / category-head emails	**Creates logins**, Watch List approver routing, Tool 2 / GRC split	single
`grc`	GRC — Buying History	Goods received by Brand × GM × colour per month / week	GRC Back Testing only	own slot

Bold = mandatory: Tool 1 and SAP `/decisions` refuse to decide until all six of F1, F2, F3, F4, Universe Master and BSR are present (`Tool1Router.MissingRequired`).

2.2 The upload paths

Path (portal)	API	What it does
A file per slot	`IngestFile` (`Key`, `Append`)	Reads one dataset; BSR appends unless `Append=false`
📦 ZIP bundle	`IngestZip`	Recognises files by name (`F1_Stock…`, `UniverseMaster…`) or as raw exports; reports missing required files
🧩 Raw export import	`IngestRaw`	Recognises SAP/BW report layouts by their headers and converts them (stock: `Stk q + In Q` → `OP STK`; sales: `Calendar Day` / `Sales Qty` dates → `Day 1..N`; PO: `Sum of Pending Qty` → `Pending Qty`; article master; BSR)
GRC slot	`IngestGrc`	Monthly (`Cal. year` row) or weekly (`WW.YYYY`) GRC layout

All paths end in the same commit: one **upload batch** (`UPL-yyyyMMdd-nnnn`), one new version per dataset, manifest update, upload log + audit line, cache invalidation; for `catheadmap` also the SCR login sync. History keeps every version, and any older version or whole upload can be made current again.

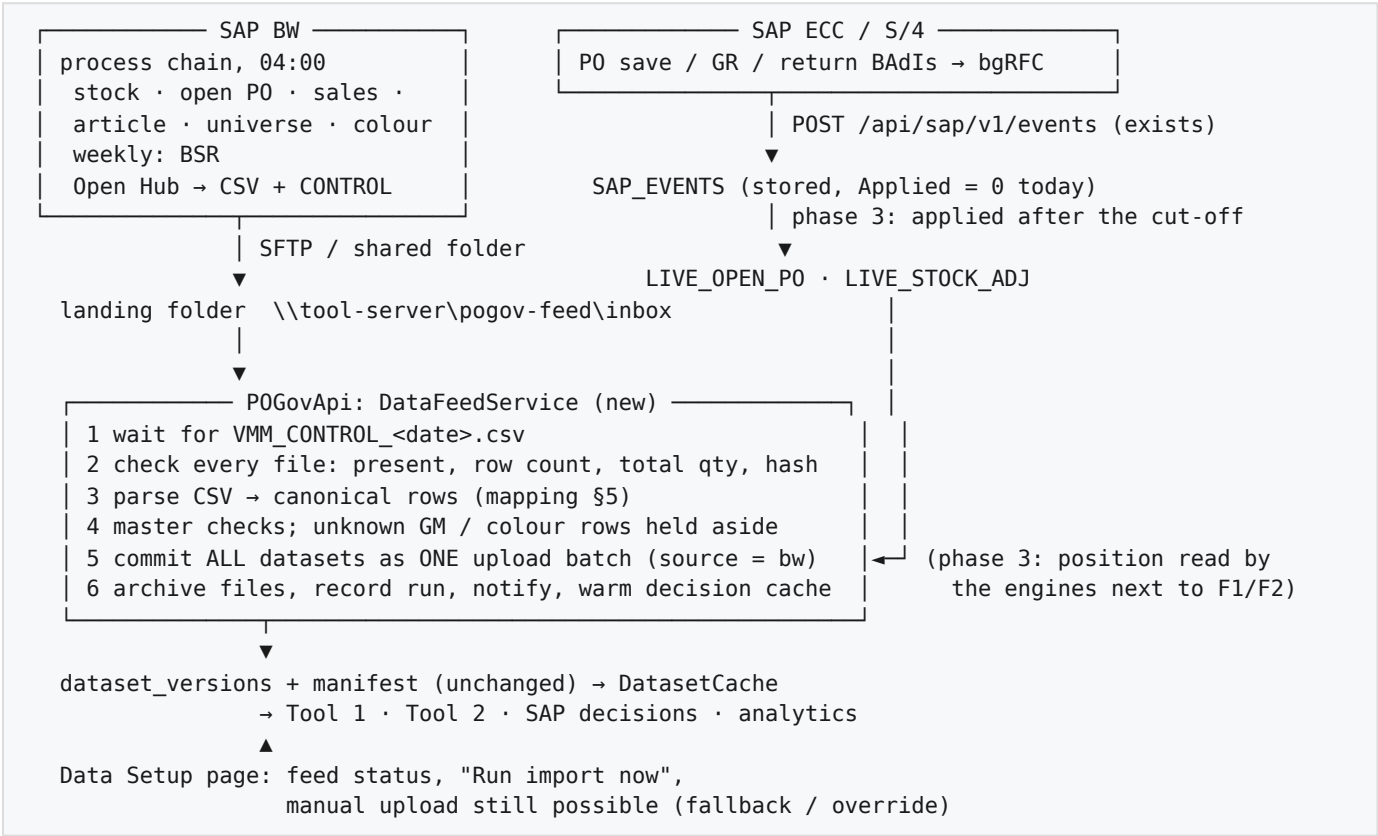
This commit is the seam for automation: the automatic import produces the same rows and calls the same `SaveVersionAsync`, so the engines, History, restore and audit work unchanged.

2.3 Problems with the manual process

- The data is only as fresh as the last person who uploaded; a missed morning means decisions on old stock and POs.
- SAP `/decisions` already answers in real time, but on files that can be a day or more old (`bw_snapshot` = upload time of F1, not the real stock date).
- Large files are exported, renamed and uploaded by hand; mistakes (wrong file in a slot, old export) are only noticed by eye.
- BSR files accumulate per upload — an old season file can stay mixed in.
- No completeness check: a half-exported file loads as long as it has headers.

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3. Target architecture



Why files from BW, not a live connection to BW: this is the delivery agreed in the Developer Brief (§04) and the least coupling: BW owns the extraction logic and the cut-off; the tool needs no BW user, no RFC library and no network path into BW; a failed night is visible as missing files. The alternatives are in §4.

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4. Delivery options considered

Option	How	For	Against	Verdict
A. BW Open Hub → CSV + control file (SFTP or shared folder)	BW process chain ends with an Open Hub Destination per dataset, writes CSV into the landing folder, control file last	Agreed in the Developer Brief; simple; BW keeps the business logic and the cut-off; files are an audit trail; replayable	One-day-old data until phase 3	**Recommended**

Option	How	For	Against	Verdict
B. Tool pulls BW queries (OData / BW query via HTTP)	Scheduled job in the tool calls published BW queries	No files to manage	Needs a BW technical user, query performance at night, paging of millions of rows, harder to reconcile	Fallback if Open Hub is not available
C. Direct database read (BW on HANA calculation views via ODBC)	SQL reads	Fast	Tight coupling to BW internals; needs HANA client and DB user; security review	Not recommended
D. SAP RFC / BAPI from the tool	NCo / RFC calls for stock, POs	Real time	Heavy (SAP NCo licence/library), puts query load on ECC, duplicates BW logic	Not recommended — SAP events cover the real-time part
E. Keep manual upload, improve it	Better ZIP rules	No project	Does not solve freshness	Stays only as fallback

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5. Dataset by dataset: source, schedule and mapping

The file names and column lists below are the ones in the Developer Brief (`VMM_<DATASET>_<YYYYMMDD>.csv`). The **right-hand column** is what the tool's code actually reads today, so the importer must produce exactly those headers (or the engines must be taught the new names — §9.4 recommends mapping in the importer).

5.1 Overview

#	Tool dataset	File	Source	Frequency	Phase
1	`f1` Stock	`VMM_STOCK`	BW (closing stock, company level)	daily 04:00	1
2	`f2` Pending PO	`VMM_OPENPO`	BW (released vendor POs, open qty)	daily 04:00	1
3	`f3` Sales Trend	`VMM_SALES`	BW (POS sales, last 45 days)	daily 04:00	1
4	`um` Universe Master	`VMM_UNIVERSE`	BW	daily 04:00	1
5	`artmaster` Article Master	`VMM_ARTICLE`	BW	daily 04:00	1
6	`colmaster` Colour Master	`VMM_COLOUR`	SAP (via BW)	daily 04:00	1
7	`bsr` BSR	`VMM_BSR`	BW	weekly (e.g. Monday 04:00)	1
8	`f4` Forecast / plan	`VMM_PLAN`	SAP SFCC / SFCR — manual upload until then	when changed (later hourly)	2
9	`attrrel` Attr Relevance	—	Planning (manual)	when changed	stays manual
10	`catheadmap` SCR mapping	—	Merchandising (manual) — or HR/merch system later	when changed	stays manual (option: phase 2)
11	`grc` GRC buying history	`VMM_GRC` (optional)	BW goods receipts	weekly	2 (optional)
—	Live PO changes, GR, returns	`POST /events`	SAP R/3	real time	3

Fixed rules from the Developer Brief that every file follows: **company level** (all sites summed), **vendor POs only (no STO)**, open PO = **released** POs only, open qty = ordered − received + returned, **no real-time sales**.

5.2 `VMM_STOCK` → `f1`

BW file column	Tool column (`f1`)	Rule
Generic Material	`Generic Material`	leading zeros removed
Colour	`Color`	colour **code** (as in Colour Master)
Brand Type	`Brand Type`	
Division	`DIV`	
MC CD	`MC CD`	
MRP	`MRP`	number, no thousand separator
Flag	`BOGO/NON-BOGO`	`BOGO` / `NON-BOGO` exactly as in F4 / Universe
Stock Qty	`OP STK`	closing stock **incl. in-transit** (today's raw export adds `Stk q + In Q`) — **confirm**
Article, Stock Value, As-of Date	—	stored in meta / checked; As-of Date must equal the control-file snapshot date
(not in brief) MC Desc	`MC DESC`	**add** — used by Attribute Analysis

5.3 `VMM_OPENPO` → `f2`

BW file column	Tool column (`f2`)	Rule
PO Number	`PO Number`	
PO Item	—	kept in meta; Tool 2 works per PO line = PO + GM + colour + month
Generic Material, Colour, Brand Type, MC CD, MRP, Flag	`Generic Material`, `Color`, `Brand Type`, `MC CD`, `MRP`, `BOGO/NON-BOGO`	as F1
Delivery Date (YYYY-MM-DD)	`Delivery Month`	converted to `MM.YYYY` (first schedule line)
Pending Qty	`Pending Qty`	= ordered − received + returned (BW computes)
Vendor	`Vendor Name`	the tool shows the **name**; brief lists only `Vendor` → **add Vendor Name**
(not in brief) PO Value	`PO Value`	**add** — Tool 2 "₹ Value at Risk" needs it
(not in brief) Division, MC Desc, Season, Art Desc	`DIV`, `MC DESC`, `Season`, `ART DESC`	**add** (resolve map fallback when a GM is not in Article Master)
Ordered / Received / Returned Qty, Release Date	—	meta; used for checks and phase 3

5.4 `VMM_SALES` → `f3`

The brief delivers **long format** (one row per GM × colour × day); the tool reads **wide format** (`Day 1` … `Day 45`, oldest first). The importer pivots:

BW file column	Tool column (`f3`)	Rule
Generic Material, Colour, Brand Type, MC CD, MRP, Flag	same as F1	one output row per GM × colour

BW file column	Tool column (`f3`)	Rule
Date + Sales Qty	`Day 1` … `Day N`	N = `params.mixWindowDays` (45); `Day 1` = snapshot date − N + 1; missing days = 0

Volume note: long format is ~45 × the wide row count. Alternative to agree with BW: deliver wide (`Day 1..45`) directly.

5.5 `VMM_UNIVERSE` → `um`

BW file column	Tool column	Rule
Generic Material	`Generic Material`	presence = Single Concept
MC CD, MRP, Flag	`MC CD`, `MRP`, `Flag (BOGO/NON-BOGO)`	
Concept (Single), Valid From	—	only rows with Concept = Single and Valid From ≤ snapshot date are loaded
(not in brief) Brand Type, Division, MC Desc	`Brand Type`, `Division`, `MC DES`	**add** (resolve map)

5.6 `VMM_ARTICLE` → `artmaster`

BW file column	Tool column	Rule
Generic Material	`Generic Material`	one row per GM (article/colour rows collapsed)
Brand Type, Division, MC CD, MC Desc, MRP, Flag, Season	`Brand Type`, `Division`, `MC CD`, `MC DESC`, `MRP`, `Flag (BOGO/NON-BOGO)`, `Season`	Season drives Current / Fallback season selection
Style, Fit, Product Material, Design	same names	Recommendation dimensions, Attribute Analysis
Vendor, Vendor Name	`Vendor`, `VENDOR NAME`	
(not in brief) Article description	`ART DESC`	**add**

5.7 `VMM_COLOUR` → `colmaster`

BW file column	Tool column	Rule
Colour Code	`Colour`	the tool's lookup reads `Colour/Color` and `Colour Desc/Color Desc/Description` — **the brief's names do not match**, the importer must rename
Colour Description	`Colour Desc`	

5.8 `VMM_BSR` → `bsr` (weekly)

BW file column	Tool column	Rule
Generic Material, Colour, MC CD, MRP, Flag	`Generic Material`, `Color`, `MC CD`, `MRP`, `Article Type`	
Weeks on Floor	`WEEK ON FLOOR`	
Avg Per-Day Sell-Thru %	`AVG PERDAY SELL THRU%`	percent number (e.g. `5.2` → stored as `5.2%`)
Season, Snapshot Week	—	the file holds **all seasons**; season of a GM still comes from Article Master

Change of behaviour: today BSR **accumulates** over uploads; the feed **replaces** BSR with the weekly file (one complete all-seasons file). Days without a new BSR file keep the current BSR version.

5.9 `VMM_PLAN` → `f4` (phase 2)

File column	Tool column	Rule
MC CD, MRP, Flag (+ Brand Type — **add**)	`MC CD`, `MRP`, `BOGO/NON-BOGO`, `Brand Type`	one row per Brand × MC × MRP × Flag
Month + Plan Qty	`MM.YYYY` columns	pivot long → wide
Plan Version	—	load only when the version changed

Until SFCC/SFCR is connected, the planner keeps uploading F4 by hand; the feed never overwrites a manually uploaded F4 with an older plan version.

5.10 What stays manual

- **Attr Relevance Master** and **SCR Products / Cat Head Mapping**: business-maintained, change rarely; keep the Data Setup slots. SCR upload keeps creating logins.
- **Parameters** and **Season Selection** (Data Setup sidebar).
- Manual upload of **any** dataset stays possible as an override (§8.4).

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6. File contract (to agree with the BW team)

Item	Rule
Location	`\\<tool-server>\pogov-feed\inbox` (shared folder) **or** SFTP account `pogov_feed` → same folder. Write-only for BW, read/move for the tool service account
File names	`VMM_<DATASET>_<YYYYMMDD>.csv`, date = **snapshot (business) date**, e.g. `VMM_STOCK_20261026.csv`
Encoding	UTF-8 (BOM optional), comma separator, `"` quoting, first row = header, **fixed column order**
Numbers	dot decimal, no thousand separators, no currency signs; empty = 0 only where stated
Dates	`YYYY-MM-DD`; timestamps ISO 8601 with offset
Codes	GM and article **without** leading zeros (or the importer strips them); colour = Colour Master code
Writing	write to `*.tmp`, rename to `.csv` when complete; **control file last**
Control file	`VMM_CONTROL_<YYYYMMDD>.csv`: `File Name, Row Count, Total Qty, Snapshot Date, Extraction Time` — one line per data file of that night (`Total Qty` = sum of the quantity column: Stock Qty / Pending Qty / Sales Qty / Plan Qty; empty for masters)
Cut-off	`Extraction Time` = the moment the snapshot represents (e.g. `2026-10-26T23:59:59+05:30`); becomes `bw_snapshot` in SAP answers and the cut-off for phase 3
Re-delivery	BW may re-deliver a night by writing the same names again with a new control file; the tool re-imports
Weekly BSR	listed in the control file only on BSR days

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7. Import process (DataFeedService)

A new background service in POGovApi, following the pattern of the existing `BackupBackgroundService` (polls every few minutes, acts once per business day after a configured hour, uses the business time zone `Asia/Kolkata`).

7.1 Schedule

Setting	Default	Meaning
`DataFeed:Enabled`	`false`	master switch (off until go-live)

Setting	Default	Meaning
`DataFeed:InboxDir` / `ArchiveDir` / `ErrorDir`	`D:\pogov-feed\inbox` …	folders
`DataFeed:StartAt`	`04:15`	start looking for today's control file
`DataFeed:LateAfter`	`06:30`	no control file by then → "feed late" alert
`DataFeed:GiveUpAt`	`09:00`	stop waiting; decisions keep yesterday's base (alert sent)
`DataFeed:PollMinutes`	`5`	check interval
`DataFeed:MaxAgeDays`	`1`	snapshot older than this → `data_delayed` warning on SAP answers (optional)
`DataFeed:AlertTo`	BW team, planning, IT	email recipients

7.2 Steps of one run

1. **Detect.** `VMM_CONTROL_<date>.csv` for a date newer than the last successful import appears.
2. **Completeness.** Every file named in the control file exists and is complete (no `.tmp`); unknown extra files are ignored and reported.
3. **Reconcile.** For each file: row count = control `Row Count`; sum of the quantity column = `Total Qty` (to the unit); SHA-256 recorded. Any mismatch → **run fails** (step 9).
4. **Parse & map.** Read the CSV with the existing delimited reader (UTF-8 path), map columns per §5 into the tool's canonical headers, convert dates (`Delivery Date` → `MM.YYYY`), pivot sales and plan to wide format, strip leading zeros.
5. **Masters first.** Load Colour, Article and Universe masters into the new snapshot before the transactional files.
6. **Reference checks.** F1 / F2 / F3 / BSR rows whose GM is in no master, or whose colour is not in Colour Master, are **held aside** (not loaded) and listed in `DATA_FEED_REJECT`; a run fails only if held-aside rows exceed a threshold (`DataFeed:MaxRejectPct`, default 2 %).
7. **Plausibility.** Compare with the current base: total stock, total open PO qty and row counts within ±`MaxSwingPct` (default 30 %) of yesterday, else **warning** (not failure) for planning to look at.
8. **Commit (one transaction per run).** One upload batch `UPL-…` with source **`bw-feed`** and the control-file date as label; one new version per dataset (same `SaveVersionAsync` as manual uploads); manifest updated once; BSR **replaced** (not appended) when a BSR file is present; F4 only when the plan version changed. `bw_snapshot` = control `Extraction Time` (stored in the dataset meta of F1, read by `SapDecisionService.BwSnapshot`). Cache invalidated; `DecisionWarmUpService` loads the new snapshot within 30 s.
9. **On failure.** Nothing is committed; the **previous base stays current**; files move to `ErrorDir/<date>/`; the run is recorded with the reason; alert email to `AlertTo`; the Data Setup page shows the red feed status. Manual upload or "Run import now" after BW re-delivers.
10. **Archive & record.** Files move to `ArchiveDir/<date>/` (kept `ArchiveKeepDays`, default 60); run, files and counts recorded; audit line `📂 Upload` with "BW feed".

7.3 Idempotency and replays

- A control-file date is imported once; re-delivery (new control file, same date) imports again and replaces that night's versions (new upload batch, old one stays in History).
- An older date than the current base is never imported automatically (admin can force it from the portal).
- Running two imports at once is prevented by the store lock already used for uploads.

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8. Changes in the tool

8.1 Database (new script `06_data_feed.sql`, schema v5)

Table	Columns (main)	Purpose
`DATA_FEED_RUN`	Run_Id, Snapshot_Date, Started_At, Finished_At, Status (RUNNING / OK / WARNING / FAILED / LATE), Reason, Upload_Id, Extraction_Time, Triggered_By (schedule / admin)	one row per import attempt
`DATA_FEED_FILE`	Run_Id, File_Name, Dataset_Key, Rows_Expected, Rows_Read, Qty_Expected, Qty_Read, Sha256, Status, Message	per file
`DATA_FEED_REJECT`	Run_Id, Dataset_Key, Line_No, Reason, Row_Json	held-aside rows (kept 60 days)

Procedures `PRO_DATA_FEED_RUN` (`START` / `FILE` / `FINISH` / `LIST` / `GET`) and `PRO_DATA_FEED_REJECT`, QPS style (`@Cmd`, `@Error_Code`, `@Error_Message`), like the SAP procedures. `upload_batches.rec.source` gets the value `bw-feed`; nothing else in the existing tables changes.

8.2 Code

Where	What
`Application/Services/V1/DataFeed/`	`DataFeedService` (steps 1–10), `FeedFileContract` (expected columns per file), `FeedMapper` (BW columns → canonical rows, pivots), `FeedValidator`
`Application/Interfaces/V1/DataFeed/`	`IDataFeedRepository`, `IFeedSource` (folder today; SFTP or BW-pull later behind the same interface)
`Infrastructure/Services/V1/DataFeedBackgroundService.cs`	schedule (as `BackupBackgroundService`)
`Infrastructure/Repositories/DataFeedSqlServerRepository.cs`	the procedures
`Infrastructure/Services/V1/Mail/`	SMTP alert sender (`Smtp` settings) — the tool has no mail sender today
`Application/Services/V1/Ingest/DataIngestService`	unchanged; the feed reuses `CommitAsync` / `SaveVersionAsync`
`SapDecisionService.BwSnapshot`	prefer `meta.extractionTime` of F1 over the upload time
`POGovApi/Controllers/V1/DataFeedController.cs`	`DataFeedStatus`, `DataFeedRuns`, `DataFeedRunGet`, `DataFeedRunNow`, `DataFeedRejectsExport` (staff; run-now admin) + API_MASTER rows

8.3 Portal

- 📂 **Data Setup**: a new top card **"Automatic data feed"** — last run (date, status, snapshot time, rows per dataset), next expected run, **Run import now** (admin), link to the run details and the held-aside rows (⬇ Excel). Each slot shows its source: "BW feed · 26 Oct snapshot" or "Manual upload by … (overrides the feed until the next run)".
- 🗂 **History → Data uploads**: type **"BW feed"** next to ZIP bundle / Raw import / File upload; make-current works as today.
- Header pill: amber when today's feed is late or failed ("Data from 25 Oct").
- SAP Interface → Overview: `bw_snapshot` shown with its source (feed / upload).

8.4 Manual upload after go-live

Kept for: emergencies (feed failed), F4 until phase 2, Attr Relevance and SCR. A manual upload of a fed dataset becomes current until the next successful feed run, which replaces it again (the slot says so).

Optionally (`DataFeed:LockFedSlots`) the fed slots can be hidden for planners and left to admins only.

8.5 Fixes to make on the way (found while preparing this document)

- Upload size limit is read from three different keys (`Ingest:MaxUploadMb`, `max_upload_mb`, `PoGovSettings:MaxUploadMb`) and hard-coded at 400 MB in the portal — unify on `PoGovSettings:MaxUploadMb`.
- Colour Master lookup does not accept `Colour Code` / `Colour Name` headers (the test fixture uses them) — accept them or document the required headers.
- ZIP and Raw imports **replace** BSR while slot uploads **append** — make the behaviour explicit in the UI.

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9. Phase 3 — live open-PO and stock position from SAP events

SAP already sends PO, goods-receipt and return events (`POST /api/sap/v1/events`, stored in `SAP_EVENTS` with `Applied = 0`). Phase 3 applies them so that decisions see POs raised, changed, received or returned **today**, and the same-day Tool 1 ledger can be switched off (handover §8 item 9).

9.1 Model

```
live open PO (per PO item)   = BW open PO (f2 base, snapshot T)
                               + PO_CREATED / PO_CHANGED / PO_RELEASED after T   (full current state
per PO)
                               − PO_CLOSED / PO_DELETED / RELEASE_REVOKED after T
                               − GR_POSTED  + GR_REVERSED                          (received)
                               + RETURN_POSTED − RETURN_REVERSED                   (only if a return
re-opens the qty — §12 Q1)
live stock adjustment        = + GR_POSTED − GR_REVERSED − RETURN_POSTED + RETURN_REVERSED   (after
T)
```

- Only events with `occurred_at` **after the BW cut-off T** (control-file extraction time) are applied — this prevents double counting (spec v1.4 p.4).
- Each new BW base clears the previous day's adjustments and re-applies the events of the new day after T.
- PO events carry the **full PO state**, so the newest state per PO replaces older ones (order by `seq`).

9.2 Changes

Item	Change
Tables	`LIVE_OPEN_PO` (PO, item, GM, colour, delivery month, open qty, last seq, last event), `LIVE_STOCK_ADJ` (GM, colour, qty delta, last seq)
Applier	background service applying `ACCEPTED` events in `seq` order every few seconds, setting `Applied = 1` / `Applied_At`
Engines	Tool 1 B / SAP decisions read open PO = live open PO instead of F2 + ledger; stock = F1 + live adjustment. Behind a switch `LivePosition:Enabled`
`data_delayed`	already computed from the event lag — becomes meaningful
Ledger	same-day Tool 1 ledger off when live position is on (Watch List approvals in the tool stay until the SAP workflow takes them over)

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10. Testing, cut-over and rollback

1. **File contract test** — BW delivers one sample night to a test folder; the importer runs in **dry-run** mode (`DataFeed:DryRun = true`: validates and reports, commits nothing).

2. **Parallel run, 2–3 weeks** — feed runs on the staging server every night; planners keep uploading by hand in production. A daily **reconciliation report** compares feed vs manual per dataset: row counts, total qty, and the **Tool 2 decision mix** and a sample of Tool 1 decisions on both bases.
3. **Go-live criteria** — 10 consecutive nights with: feed complete before 06:30, control checks passing, held-aside rows < 0.5 %, Tool 2 decision differences explained.
4. **Go-live** — `DataFeed:Enabled = true` in production; planners stop daily uploads (slots remain).
5. **Rollback** — switch the feed off; manual upload works exactly as today; History lets admins make any previous upload current again.

Automated tests: mapping per file (CSV fixtures for every `VMM_*` file), pivot of sales / plan, control-file mismatch, held-aside rows, idempotent re-delivery, BSR replace, F4 version skip, failure keeps the previous base, SQL procedures (SQL Server test suite).

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11. Plan, effort and responsibilities

Phase	Scope	Tool team	BW / SAP team	Duration
0	Agree file contract (§6), answer §12	2–3 days	Open Hub design	1–2 weeks
1	Importer for STOCK, OPENPO, SALES, UNIVERSE, ARTICLE, COLOUR, BSR; run log; alerts; Data Setup feed card; dry run	~4 weeks build + test	Open Hub destinations, process chain, folder/SFTP	5–6 weeks incl. parallel run
2	PLAN from SFCC/SFCR; optional GRC file; SCR from a system source (if available)	~2 weeks	SFCC/SFCR extract	2–3 weeks
3	Live open-PO / stock position from SAP events; ledger off	~5–6 weeks	confirm event rules (§12), bgRFC in production	6–8 weeks

Who	Responsible for
BW team	extraction logic, Open Hub, process chain timing, control file, re-delivery
SAP team	events (already built), GRT rule, PO workflow
Planning	parallel-run review, F4 until phase 2, Attr Relevance, parameters
Merchandising	SCR mapping
IT infrastructure	landing folder / SFTP, service account, SMTP relay, folder backup
Tool team	importer, run log, portal, alerts, tests, reconciliation report

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12. Open questions (need answers before build)

#	Question	Owner	Why it matters
1	Does a goods return (122 / 161) **re-open** the pending PO quantity? (handover §8 item 6)	SAP + Planning	open-PO formula, phase 3
2	Exact extraction list and **cut-off time**; POS posting time vs 04:00 (handover §8 item 7)	BW	`bw_snapshot`, double counting
3	Does `Stock Qty` include **in-transit**? (today's raw export adds `Stk q + In Q`)	BW + Planning	F1 = what the engines treat as stock
4	Can `VMM_OPENPO` add **PO Value, Vendor Name, Division, MC Desc, Season, Art Desc**; `VMM_STOCK` add **MC Desc**; `VMM_UNIVERSE` add **Brand Type, Division, MC Desc**; `VMM_ARTICLE` add **Art Desc**?	BW	fields the tool uses today (§5)
5	Sales file **wide** (`Day 1..45`) or long?	BW	volume, pivot effort
6	BSR day of the week; is it always all seasons in one file?	BW	BSR replace logic
7	Folder vs SFTP; server path; who monitors	IT	§6
8	Who receives failure emails; is an SMTP relay available?	IT + Planning	alerts

#	Question	Owner	Why it matters
9	Plan (SFCC/SFCR): version field, frequency, Brand Type in the key	Planning + SAP	phase 2
10	Should manual upload of fed datasets stay open to planners after go-live?	Planning	§8.4
11	Approximate row counts per file (stock, open PO, sales)	BW	sizing; tool currently tested with 400 MB uploads

Appendix A — Columns the tool reads today, per dataset (exact headers)

Header matching for most engines ignores case and spaces (`SheetRow.Col`); the Recommendation, Attribute Analysis, GRC and Cat-Head lookups need the **exact** header — the importer therefore always writes the first name listed.

Dataset	Columns (first name = the one the importer writes; alternates accepted today)
f1	`Generic Material` · `Color` (Colour) · `Brand Type` · `DIV` (Division) · `MC CD` · `MC DESC` · `MRP` · `BOGO/NON-BOGO` (Flag, Article Type, Flag (BOGO/NON-BOGO)) · `OP STK` (Op Stk)
f2	`PO Number` (PO No, Purchase Order …) · `Generic Material` · `Color` · `Brand Type` · `DIV` · `MC CD` · `MC DESC` · `MRP` · `BOGO/NON-BOGO` · `Delivery Month` (`MM.YYYY`; DEL MONTH) · `Pending Qty` (Released Qty) · `PO Value` (PO Val, Value, PO Amount) · `Vendor Name` (Vendor) · `Season` · `ART DESC`
f3	`Generic Material` · `Color` · `Brand Type` · `MC CD` · `MRP` · `BOGO/NON-BOGO` · `Day 1` … `Day N` (`D N`; oldest first) — or `MM.YYYY` month columns / `CY APR & MAY AVG`
f4	`Brand Type` · `MC CD` · `MRP` · `BOGO/NON-BOGO` · `MM.YYYY` month columns
um	`Generic Material` · `Brand Type` · `Division` · `MC CD` · `MC DES` · `MRP` · `Flag (BOGO/NON-BOGO)`
bsr	`Generic Material` · `Color` · `MC CD` · `MRP` · `Article Type` (BOGO/NON-BOGO, Flag) · `WEEK ON FLOOR` (WOF) · `AVG PERDAY SELL THRU%`
artmaster	`Generic Material` · `ART DESC` · `Brand Type` · `Division` · `MC CD` · `MC DESC` · `MRP` · `Flag (BOGO/NON-BOGO)` · `Season` · `Style` · `Fit` · `Product Material` · `Design` · `Vendor` · `VENDOR NAME` (any other column = an attribute in Attribute Analysis)
colmaster	`Colour` (Color) · `Colour Desc` (Color Desc, Description, Desc)
attrrel	`Brand Type` · `MC CD` · `MRP` · `BOGO/NON-BOGO` · `Assessment Dimension`
catheadmap	`Brand Type` · `MC CD` · `MRP` · buyer / category-head email columns (Level 1/2 → buyer, Level 3 → category head; `Category_Email_ID` for Tool 2 / GRC)
grc	own layout: `Brand Type`, `Generic Material`, art description, `Colour`; monthly `MM.YYYY` under a `Cal. year` row or weekly `WW.YYYY` with `GRC Q` / `GRC V`

Formats: delivery / plan months `MM.YYYY`; day columns `Day N`; numbers as plain numbers; per-day sell-through as a percentage.

Appendix B — Sources

- `legacy/python-app/docs/PO_Governance_Developer_Brief_Stage1_2_3.pdf` §04 (data sources, Open Hub, file list, server steps), `PO_Governance_SAP_API_Specification_v1.4.pdf` (cut-off rule, SAP fields), `DEVELOPER-HANDOVER.md` §5 and §8 (future sources, open items 6, 7, 9), `PO_Governance_Functional_Logic_Guide.pdf` p.2 (files).
- Code: `DecisionEngine/Model/DatasetKeys.cs`, `Application/Services/V1/Ingest/LegacyImport.cs`, `DataIngestService.cs`, `Application/Services/V1/Platform/DataService.cs`, `Infrastructure/Services/V1/DatasetCache.cs`, `BackupService.cs`, `database/sqlserver/05_sap_interface.sql` (`SAP_EVENTS`).